Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 14x + 2$

(2) Find x-intercepts of $y = x^2 - 25$

(3) Find x-intercepts of $y = x^2 - 4$

(4) $y = -x^2$ opens:

(5) Find x-intercepts of $y = x^2 - 9$

(6) Find vertex of $y = x^2 + 14x + 1$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 8x + 2$

(2) Find vertex of $y = x^2 + 10x + 4$

(3) $y = -2x^2$ opens:

(4) Find x-intercepts of $y = x^2 - 9$

(5) Find x-intercepts of $y = x^2 - 25$

(6) Find x-intercepts of $y = x^2 - 9$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 12x + 8$

(2) Find vertex of $y = x^2 + 14x + 8$

(3) Find x-intercepts of $y = x^2 - 9$

(4) Find vertex of $y = x^2 + 4x + 2$

(5) Find vertex of $y = x^2 + 16x + 8$

(6) Find vertex of $y = x^2 + 4x + 9$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 25$

(2) $y = -3x^2$ opens:

(3) $y = 3x^2$ opens:

(4) Find x-intercepts of $y = x^2 - 9$

(5) Find x-intercepts of $y = x^2 - 4$

(6) $y = 3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 14x + 2$

(2) Find x-intercepts of $y = x^2 - 9$

(3) Find x-intercepts of $y = x^2 - 16$

(4) Find x-intercepts of $y = x^2 - 25$

(5) Find vertex of $y = x^2 + 4x + 3$

(6) Find vertex of $y = x^2 + 16x + 5$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 16$

(2) $y = -3x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 16$

(4) Find vertex of $y = x^2 + 4x + 7$

(5) Find x-intercepts of $y = x^2 - 25$

(6) Find x-intercepts of $y = x^2 - 25$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 2x^2$ opens:

(2) Find x-intercepts of $y = x^2 - 9$

(3) Find vertex of $y = x^2 + 6x + 1$

(4) Find vertex of $y = x^2 + 4x + 7$

(5) $y = 2x^2$ opens:

(6) $y = 2x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 16$

(2) Find x-intercepts of $y = x^2 - 4$

(3) Find vertex of $y = x^2 + 16x + 1$

(4) Find vertex of $y = x^2 + 16x + 3$

(5) $y = -2x^2$ opens:

(6) Find x-intercepts of $y = x^2 - 16$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 12x + 7$

(2) Find x-intercepts of $y = x^2 - 4$

(3) $y = -2x^2$ opens:

(4) Find x-intercepts of $y = x^2 - 25$

(5) Find vertex of $y = x^2 + 10x + 8$

(6) Find vertex of $y = x^2 + 4x + 5$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 3x^2$ opens:

(2) Find vertex of $y = x^2 + 10x + 4$

(3) Find x-intercepts of $y = x^2 - 16$

(4) $y = 2x^2$ opens:

(5) Find vertex of $y = x^2 + 16x + 9$

(6) $y = -3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 14x + 3$

(2) $y = 2x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 9$

(4) Find vertex of $y = x^2 + 16x + 8$

(5) Find x-intercepts of $y = x^2 - 9$

(6) Find x-intercepts of $y = x^2 - 16$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -x^2$ opens:

(2) Find vertex of $y = x^2 + 14x + 9$

(3) $y = 3x^2$ opens:

(4) $y = 3x^2$ opens:

(5) Find vertex of $y = x^2 + 14x + 6$

(6) $y = 3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 12x + 4$

(2) $y = 2x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 9$

(4) Find vertex of $y = x^2 + 10x + 8$

(5) $y = 2x^2$ opens:

(6) Find vertex of $y = x^2 + 16x + 8$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -x^2$ opens:

(2) $y = 3x^2$ opens:

(3) Find vertex of $y = x^2 + 14x + 8$

(4) Find x-intercepts of $y = x^2 - 16$

(5) Find vertex of $y = x^2 + 16x + 4$

(6) Find x-intercepts of $y = x^2 - 16$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 3x^2$ opens:

(2) $y = 2x^2$ opens:

(3) $y = -x^2$ opens:

(4) $y = 2x^2$ opens:

(5) Find vertex of $y = x^2 + 4x + 4$

(6) Find vertex of $y = x^2 + 8x + 6$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find x-intercepts of $y = x^2 - 16$

(2) Find vertex of $y = x^2 + 10x + 1$

(3) Find vertex of $y = x^2 + 10x + 7$

(4) Find vertex of $y = x^2 + 12x + 9$

(5) Find vertex of $y = x^2 + 16x + 2$

(6) $y = 3x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -x^2$ opens:

(2) $y = 3x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 4$

(4) Find vertex of $y = x^2 + 6x + 8$

(5) Find vertex of $y = x^2 + 16x + 7$

(6) Find vertex of $y = x^2 + 8x + 4$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = 3x^2$ opens:

(2) $y = -2x^2$ opens:

(3) Find x-intercepts of $y = x^2 - 4$

(4) $y = -3x^2$ opens:

(5) $y = 2x^2$ opens:

(6) $y = 2x^2$ opens:

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) $y = -2x^2$ opens:

(2) $y = 3x^2$ opens:

(3) Find vertex of $y = x^2 + 14x + 3$

(4) Find vertex of $y = x^2 + 10x + 6$

(5) $y = 2x^2$ opens:

(6) Find x-intercepts of $y = x^2 - 9$

Section 9. Graphs of Quadratic Functions

Find the requested feature of each quadratic function.

(1) Find vertex of $y = x^2 + 4x + 7$

(2) Find vertex of $y = x^2 + 10x + 4$

(3) $y = -x^2$ opens:

(4) Find vertex of $y = x^2 + 16x + 7$

(5) Find vertex of $y = x^2 + 14x + 2$

(6) Find x-intercepts of $y = x^2 - 25$

F81a

(1) $(-7, -47)$
(2) $x = \pm 5$
(3) $x = \pm 2$
(4) down
(5) $x = \pm 3$
(6) $(-7, -48)$

F82a

(1) $(-4, -14)$
(2) $(-5, -21)$
(3) down
(4) $x = \pm 3$
(5) $x = \pm 5$
(6) $x = \pm 3$

F81b

(1) $x = \pm 5$
(2) down
(3) up
(4) $x = \pm 3$
(5) $x = \pm 2$
(6) up

F82b

(1) $(-6, -28)$
(2) $(-7, -41)$
(3) $x = \pm 3$
(4) $(-2, -2)$
(5) $(-8, -56)$
(6) $(-2, 5)$

F85a

(1) $(-6, -29)$
(2) $x = \pm 2$
(3) down
(4) $x = \pm 5$
(5) $(-5, -17)$
(6) $(-2, 1)$

F86a

(1) up
(2) $(-5, -21)$
(3) $x = \pm 4$
(4) up
(5) $(-8, -55)$
(6) down

F85b

(1) down
(2) $(-7, -40)$
(3) up
(4) up
(5) $(-7, -43)$
(6) up

F86b

(1) $(-7, -46)$
(2) up
(3) $x = \pm 3$
(4) $(-8, -56)$
(5) $x = \pm 3$
(6) $x = \pm 4$

F87a

(1) $(-6, -32)$
(2) up
(3) $x = \pm 3$
(4) $(-5, -17)$
(5) up
(6) $(-8, -56)$

F88a

(1) down
(2) up
(3) $(-7, -41)$
(4) $x = \pm 4$
(5) $(-8, -60)$
(6) $x = \pm 4$

F87b

(1) $x = \pm 4$
(2) $(-5, -24)$
(3) $(-5, -18)$
(4) $(-6, -27)$
(5) $(-8, -62)$
(6) up

F88b

(1) up
(2) up
(3) down
(4) up
(5) $(-2, 0)$
(6) $(-4, -10)$

F83a

(1) $(-7, -47)$
(2) $x = \pm 3$
(3) $x = \pm 4$
(4) $x = \pm 5$
(5) $(-2, -1)$
(6) $(-8, -59)$

F84a

(1) $x = \pm 4$
(2) down
(3) $x = \pm 4$
(4) $(-2, 3)$
(5) $x = \pm 5$
(6) $x = \pm 5$

F83b

(1) $x = \pm 4$
(2) $x = \pm 2$
(3) $(-8, -63)$
(4) $(-8, -61)$
(5) down
(6) $x = \pm 4$

F84b

(1) up
(2) $x = \pm 3$
(3) $(-3, -8)$
(4) $(-2, 3)$
(5) up
(6) up

F89a

(1) down
(2) up
(3) $x = \pm 2$
(4) $(-3, -1)$
(5) $(-8, -57)$
(6) $(-4, -12)$

F90a

(1) up
(2) down
(3) $x = \pm 2$
(4) down
(5) up
(6) up

F89b

(1) $(-2, 3)$
(2) $(-5, -21)$
(3) down
(4) $(-8, -57)$
(5) $(-7, -47)$
(6) $x = \pm 5$

F90b

(1) down
(2) up
(3) $(-7, -46)$
(4) $(-5, -19)$
(5) up
(6) $x = \pm 3$